

Karunadu Technologies Private Limited

Internship Project



Project Title

Industrial–Residential Air Quality Classification using Machine Learning & Django



Introduction

Movie analytics and recommendation systems are widely used in streaming platforms such as Netflix, Amazon Prime, and movie review platforms. Machine Learning can be used to analyze movie metadata such as genre, runtime, language, ratings, and user engagement to predict movie popularity or ratings.

In this assignment, students will develop a **Machine Learning-based web application** to classify or predict **movie rating category or popularity level** based on movie features.



Dataset

The dataset contains various attributes such as:

Movie Title

Director

Genres

Runtime

Language

Studio

Watches

Likes

List Appearances

Fans

Rating Counts (Low / Medium / High)

Average Rating

Movie Description (Text data)

Note: Interns will build and compare multiple classification models to predict the output label.



Objective

- Apply Machine Learning classification algorithms
- Understand data preprocessing and feature engineering
- Build a web application using Django
- Compare performance of different algorithms

Backend

Python
Django
Framework
Scikit-learn

Frontend

HTML
CSS / Bootstrap
Django Templates

Expected Output

- User login system
- Data input form
- Disease prediction output
- Accuracy comparison of models



Tasks

Interns must implement **at least 4 classification algorithms** from the list below:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes



Tasks

Web Application (Django)

Interns must build a **full-stack web application** using Django.

Login Module

- User Registration
- Login Authentication
- Session Management

Input Form Page

Build Django UI with input form of all Inputs

Prediction Module

Add a **“Predict” button**

On submission:

- Load trained ML models
- Predict output label
- Display result on UI

Results Display

- Predicted Output
- Accuracy of each algorithm
- Best performing algorithm



Important Questions

- What is classification in Machine Learning? How is it different from regression?
- Explain the importance of the output in this dataset.
- What is feature selection? Why is it important in this project?
- Explain the difference between supervised and unsupervised learning.
- What is overfitting and underfitting? How can it be avoided?
- What is the role of training and testing datasets?
- Explain how Logistic Regression works for classification.
- How does a Decision Tree decide splits?
- What is the advantage of Random Forest over Decision Tree?
- Explain how KNN works and its limitations.
- What is the concept behind SVM (Support Vector Machine)?
- What is a Confusion Matrix? Interpret it for your model.
- What preprocessing steps did you apply to the dataset?
- How did you handle missing or inconsistent data?
- Explain the flow of your Django application.
- How does authentication (login system) work in your project?
- What is the role of views, templates, and models in Django?
- How is the prediction triggered from the frontend?



Submission Requirements

Interns must submit

- Internship Report
- Source Code (Django + ML models)
- Dataset (if modified)
- Project Report (PDF) including:
 - Problem Statement
 - Methodology
 - Algorithms Used
 - Results & Comparison
 - Screenshots



Contact US



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**Thank You Interns for being a
part of Internship Journey**

**Contact Karunadu Technologies Private Limited
For Internship**